

Action and Inaction in Moral Judgments and Decisions: Meta-Analysis of Omission Bias Omission-Commission Asymmetries

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Siu Kit Yeung¹, Tijen Yay², and Gilad Feldman¹ 

Abstract

Omission bias is people's tendency to evaluate harm done through omission as less morally wrong and less blameworthy than commission when there is harm. However, findings are inconsistent. We conducted a preregistered meta-analysis, with 21 samples (13 articles, 49 effects) on omission-commission asymmetries in judgments and decisions. We found an overall effect of $g = 0.45$, 95% confidence interval (CI) = [0.14, 0.77], with stronger effects for morality and blame than for decisions. Publication bias tests produced mixed results with some indication for publication bias, though effects persisted even after most publication bias adjustments. The small sample of studies included limited our ability to draw definite conclusions regarding moderators, with inconclusive findings when applying different models. After compensating for low power, we found indication for moderation by role responsibility, perspective (self vs. others), outcome type, and study design. We hope this meta-analysis will inspire research on this phenomenon and applications to real-life, especially given the raging pandemic. Materials, data, and code are available on <https://osf.io/9fcqm/>.

Keywords

omission bias, commission, morality, judgment and decision-making, preregistered meta-analysis

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Introduction

A large body of social and cognitive psychology research claimed empirical support for an omission bias, commonly defined as evaluating commission as more morally wrong and blameworthy compared with omission, in cases of potential harm. Omission commonly refers to no deviation from a reference point, either doing nothing, a deliberate choice of not taking action, or making no change (Feldman et al., 2020). Commission commonly refers to a deviation from a reference point, making an active decision, doing something, or making a change (Feldman et al., 2020). For example, omission bias has been demonstrated in the preference not to vaccinate (omission) rather than to vaccinate (commission; Asch et al., 1994; Ritov & Baron, 1990), or in preference for doing nothing (omission) over doing something (commission) to prevent harm that can be easily stopped (Spranca et al., 1991).

Omission bias studies demonstrated that when faced with a dilemma between taking action and doing nothing, with both likely resulting in similar negative outcomes, people tend to prefer omission over commission. Omission bias holds implications for several domains, such as moral and/or legal judgments (Spranca et al., 1991) and in medical/health

decision-making regarding vaccinations (Connolly & Reb, 2003). These seem especially relevant given the raging COVID-19 pandemic during the time of writing (the year 2021). The omission bias has been linked to the “action principle of harm” which describes a phenomenon in which harm through action tends to be judged as morally worse than harmful omission (Cushman et al., 2006).

However, several scholars raised doubts regarding the reliability, generalizability, and boundary conditions of the omission bias effect. The methods for studying the effect were criticized, such as the choice of the moral scenarios used in the early demonstrations, raising concerns about oversimplicity and relevance for the complexity of real-life situations (e.g., Connolly & Reb, 2003). Further criticism was raised regarding methodology, such as the use of numerical risk-balancing¹ procedures in some vaccination

¹The University of Hong Kong, Hong Kong

²Maastricht University, The Netherlands

Corresponding Author:

Gilad Feldman, Department of Psychology, The University of Hong Kong, Pok Fu Lam Road, Hong Kong SAR 999077.
Email: gfeldman@hku.hk

scenarios (Connolly & Reb, 2003, 2012) and the lack of control for equivalence of norms violated by commission and omission (Willemsen & Reuter, 2016). Moreover, some observed weaker effects when using between-subject designs compared with within-subject designs (e.g., Jamison et al., 2020). It has also been suggested that omission bias is sensitive to moderating factors, such as familiarity with the target, social role responsibility for the target (Haidt & Baron, 1996), outcome (Spranca et al., 1991), and the target (self vs. others; Ubel et al., 2011).

The importance and relevance of the phenomenon combined with the mixed findings raise the need for a meta-analytic review of the existing evidence. A meta-analysis allows for a systematic integration and comparison of different results to attempt more accurate estimations of the true effect. We aimed to determine the overall weighted effect size of omission-commission (action-inaction) asymmetries on judgments of morality, blame, and decisions. We also aimed to examine possible moderators of the effect and to explore potential directions that have yet to receive attention.

We first review the literature and discuss the theories and hypotheses. Then, we discuss possible moderators. After that, we report the meta-analysis we conducted and discuss its findings. Finally, we discuss uncertainties regarding the phenomenon and possible future research directions.

Scope

In this meta-analysis, we focused on omission bias as capturing omission-commission asymmetries regarding morality and blame judgments and decisions. We make the distinction between omission bias and other related biases, and attempted to clearly define the scope of the investigation in our methods to differentiate omission bias from other phenomena (see Feldman et al., 2020, for a review). We do not address omission-commission asymmetries related to emotions (action-effect), or normal-abnormal asymmetries (normality, exceptionality effect, status quo bias, default bias, etc.). We also attempted to disentangle omission bias from trolley like moral dilemmas, by focusing on studies in which the outcomes are the same or very similar, and therefore do not confound omission bias with utilitarianism versus deontology. We also note, that in the omission bias literature, the vast majority of studies adopted a narrow meaning of commission as being about “doing something” and of omission as being about “doing nothing/not doing something” (rather than about “change” or “no change” from some reference point).

Theories, Mechanisms, Explanations, and Hypotheses

Spranca et al. (1991) were among the first to discuss possible explanations for the omission bias. The causal relevance hypothesis focuses on the different causal attributions in

omission and commission (Willemsen & Reuter, 2016). Spranca et al. (1991) suggested that people’s actions were judged as having stronger causal links to the outcome compared with inactions. Follow-up studies supported this idea (Henne et al., 2019; Kordes-de & Vaal, 1996; Willemsen & Reuter, 2016) with action being perceived as more intentional than inaction (Hayashi, 2015; Jamison et al., 2020; Kordes-de Vaal, 1996). A related explanation is about the role of information. Lacking information in an uncertain situation makes people more sensitive to action cues (Frisch & Baron, 1988; Spranca et al., 1991). Ritov and Baron (1990) found more statistical information resulted in weaker omission bias, supporting this notion. With limited information, people’s judgments may be more biased by action cues, which is associated with higher causality and higher perceived intentionality mentioned above.

Omission bias is related to the classic action effect (Kahneman & Tversky, 1982), the first demonstration of action-inaction asymmetries in the domain of evaluations and emotions (Anderson, 2003; Connolly et al., 1997; Ritov & Baron, 1995). Although action-effect and omission bias are related, and omission bias may partly be explained by action-effect (discussed below), there are key differences. Omission bias focuses on omission-commission asymmetries regarding morality, blame, and decisions. Specifically, it is often mentioned in contexts of moral judgments, especially regarding harm and blame (DeScioli et al., 2012). Action-effect focuses on the emotional reactions to outcomes, most notably the demonstration of stronger regret over negative outcomes for action than inaction. Action-effect seems broader and extends beyond moral contexts, such as the classic financial investment context used in Kahneman and Tversky (1982). See Feldman et al. (2020) for a detailed discussion and comparison.

The link between omission bias and action-effect lies in the attempt to avoid stronger negative feelings associated with harmful outcomes resulting from having taken action. People may aim to prevent future regret over harm inflicted by not acting. However, action-effect likely cannot fully account for the omission bias (DeScioli et al., 2011). For example, in a recent replication and extension of the classic omission bias study by Spranca et al. (1991), Jamison et al. (2020) measured attributions of both morality and regret in moral dilemmas and found regret was more strongly associated with inaction rather than action, opposite to findings of Kahneman and Tversky’s (1982) nonmoral investor scenario. In this case, it is likely that this was because those scenarios involved intended harm, so regret was associated with having not inflicted any harm. Therefore, the link between regret and morality seems to do more with reference points, goals, and intent rather than strictly with morality.

Sense of responsibility, which is associated with but distinct from regret (for details, see Zeelenberg et al., 1998), is another key element in the moral judgments of omission and commission. Sense of responsibility is closely associated

with causal attributions and intentionality mentioned above. Kordes-de Vaal (1996) and Zeelenberg et al. (2000) both found that commissions were perceived as more causal than omissions.

Commissions also seem to be more salient and clearer to define than omissions (Kordes-de Vaal, 1996). Inaction could be a deliberate decision not to act, but it could simply reflect not having made any choice, or choosing to defer the decision. In choosing between harm through action and harm through inaction, people may strategically aim to minimize accountability, responsibility, possible blame, and possible punishment through the more ambiguous inaction rather than the clearer more observable action (termed “omission strategy”; DeScioli et al., 2011a).

Omission Bias: Main-Effect

Our first aim was to examine the effect of the omission bias. We expected the evidence to be in support of the omission bias as meaningfully different from the null (null not included in confidence intervals [CIs]). The below hypotheses are based on the assumption that the consequences of action and inaction are the same.

Hypothesis 1a (H1a): Harm through action is perceived as less moral than harm through inaction. Harm through action is perceived as more morally accountable and blameworthy than harm through inaction.

Hypothesis 1b (H1b): When facing a dilemma between inflicting undesired harm through action or inaction, harm through inaction is preferred to harm through action.

Moderators

The demonstration of omission bias was followed by studies that examined potential boundary conditions and factors that impact the strength of omission bias. We discuss each of the theorized moderators below, yet we note that after completing the coding procedure, we realized that for one of the pre-registered moderators—availability of statistical information regarding chances of harm through omission and commission—there were too few studies for a meaningful analysis. We therefore removed this moderator and do not discuss it in this section, and we detail this deviation in the Supplemental Material (subsection “Preregistration Versus Final Report Deviations”).

Familiarity With Target and Social Role Responsibility for the Target

We classified familiarity with target into two categories: (a) decision maker knows the target personally (e.g., friends, family members), and (b) decision maker does not know the target personally (e.g., stranger).

We classified social role responsibility over the target into two types: (a) having clear responsibility over target (e.g., as

a parent, as a doctor, under the social role to prevent harm or directly responsible for the negative outcome), and (b) unclear responsibility over target (e.g., as a friend, not under the direct social role to prevent harm).

The above two factors have been previously suggested as moderators of omission bias. The two are likely related, as having more responsibility over a target is likely to be associated with higher familiarity with the target. Role responsibilities over familiar targets—such as between family members—involve more concern regarding possible harm than between strangers that the person is unfamiliar with (Haidt & Baron, 1996).

Haidt and Baron (1996) found that omission bias was substantially weakened for decision makers with responsibility over the target aiming to minimize harm similarly through both commission and omission. This is due in part to social role-based expectations for caregivers to do what they can to prevent harm to their care dependents (Baron & Ritov, 2004; Haidt & Baron, 1996). However, we also note that there appears to be evidence supporting a meaningful and strong omission bias for familiar targets. For example, in their classic omission bias article, Ritov and Baron (1990) demonstrated the effect for cases of vaccination for children by their parents, with the explanations provided by parents who refuse to vaccinate, thereby exhibiting omission bias, were related to their responsibility for their children’s life or death. However, in a follow-up heated debate in the literature, Connolly and Reb (2003) offered contradictory findings and despite several back-and-forth debates between the two groups regarding methodology claiming to have resolved the issue, the mixed findings suggest complex moderators.

However, there are situations in which the decision maker does not have a close relationship with the target but is responsible for the target, such as medical doctors who are responsible for their patients’ health (e.g., Baron & Ritov, 2004; Spranca et al., 1991). These are especially relevant in pandemic situations in which caregivers, key health policy-makers, and doctors have to decide whether they vaccinate, provide medical treatments, or make key public health decisions (act or do not act).

We set forth the following hypotheses:

Hypothesis 2a (H2a): Omission bias is stronger for more familiar targets.²

Hypothesis 2b (H2b): Omission bias is stronger for less familiar targets.³

Hypothesis 3a (H3a): Omission bias is stronger with higher perceived responsibility for the target.

Hypothesis 3b (H3b): Omission bias is stronger with lower perceived responsibility for the target.

Presence/Absence of Negative Outcome

People often evaluate the quality of a decision based on the outcome (Baron & Hershey, 1988). Spranca et al. (1991) proposed that the valence of an outcome may moderate

evaluations of morality and responsibility. They found that some participants exhibited outcome bias in omission bias, with stronger omission bias when outcomes were negative and involved actual harm to the target. In some omission bias studies, there is no outcome information specified (e.g., DeScioli et al., 2012), or it is uncertain if the outcome will be negative (e.g., Connolly & Reb, 2003). In a recent replication of this work by Jamison et al. (2020), the researchers found mixed support depending on the scenario used.

We hypothesized as follows:

Hypothesis 4 (H4): There would be an omission bias effect regardless of outcome (null not included in CI).

Hypothesis 5 (H5): Omission bias is stronger when outcomes are negative compared with when outcomes are not negative (neutral/positive/unknown/uncertain).

Experimental Design

There has been some debate in the literature on action and inaction regarding study designs, on differences in effect size comparing within-subject with between-subject designs. N'gbala and Branscombe (1997) first suggested that action-effect in within-subject designs may not replicate well to between-subject designs, though since then there have been many follow-up studies that have convincingly demonstrated the action-effect also using between-subject designs, though possibly with weaker effects (e.g., Feldman & Albarracín, 2017). There has been a similar debate in the omission bias literature between the Baron and Ritov team and the Connolly and Reb team contrasting different study designs. Since this debate, there have been successful extensions of omission bias initially demonstrated using within-subject design (e.g., Spranca et al., 1991) to between-subject designs (Jamison et al., 2020). Given the mixed findings and the ensuing debates regarding both omission bias and action-effect, we aimed to examine study design as a moderator and compare the omission bias effect size using these two designs. We therefore outlined the following competing hypotheses:

Hypothesis 6a (H6a): Omission bias is stronger for studies using a within-subject design compared with between-subject designs.

Hypothesis 6b (H6b): Study design has no impact on the omission bias (overlapping CIs).

Target: Self Versus Others

We did not preregister this moderator, yet we identified this as a moderator while coding, noticing that some studies use self as the target. In medical treatments such as vaccination, psychiatric treatment, surgery, and cancer treatments, studies found that when deciding for others, people (laypeople or medical doctors) are more likely to prefer the active decision compared with when deciding for self, in which people are

more likely to exhibit omission bias (Janssen et al., 2015; Mendel et al., 2010; Ubel et al., 2011; Zikmund et al., 2006). This may be because when deciding for others, people are in the role of being responsible for others by taking action to prevent harm. Another explanation is that when deciding for others, people tend to focus on a single dimension of the decision, which is the risk-benefit tradeoff, an easily defensible justification whereas when deciding for self, people are influenced by more factors such as emotions and biases (Ubel et al., 2011). Therefore, as an exploratory hypothesis, we hypothesized the following:

Exploratory hypothesis: The omission bias effect is stronger when the target is self compared with when the target is others.

Method

Preregistration and Open Science

We preregistered the meta-analysis on the Open Science Framework before search and data coding (<https://osf.io/hyebx/>; <https://osf.io/jw62m/>; these are the same registrations, only the latter was adjusted for the “preregistration challenge” format conducted by the Center of Open Science, more details on: <https://osf.io/x5w7h/>). Preregistration, coding sheet, and all additional materials used in the meta-analysis were made available on the Open Science Framework (<https://osf.io/9fcqm/>).

We included a coding sheet with a codebook detailing all fields, elaborated, and coded, which is used directly by the accompanied Rmarkdown code and exported outputs detailing all our analyses and results (see “Coding and analyses” directory on the OSF). We report all search, coding, measures, and exclusions in this meta-analysis.

Literature Search

To find articles relevant to our topic, we used Google Scholar (for suitability for meta-analyses, see Gehanno et al., 2013; Walters, 2007) and identified a sample of studies based on various steps illustrated in Figure 1. In the first initial online search, we decided to identify articles including variations of keywords such as *omission bias*, *omission effect*, *action principle*, *commission*, and *omission*, aimed to identify relevant literature, related topics, and the scholars in this field. As a result of the first search round, we were able to identify more specific search terms on the omission bias. Consequently, our search patterns were as follows: (“*omission bias*” OR “*omission effect*” OR “*omission strategy*” OR “*action effect*” OR “*action principle*” OR [Baron AND Ritov] OR [Connolly AND Reb] OR [action AND inaction] OR [actions AND inactions] OR [omission AND commission] OR [omissions AND commissions] OR [direct AND indirect] OR [doing AND allowing] OR [active AND passive]) AND

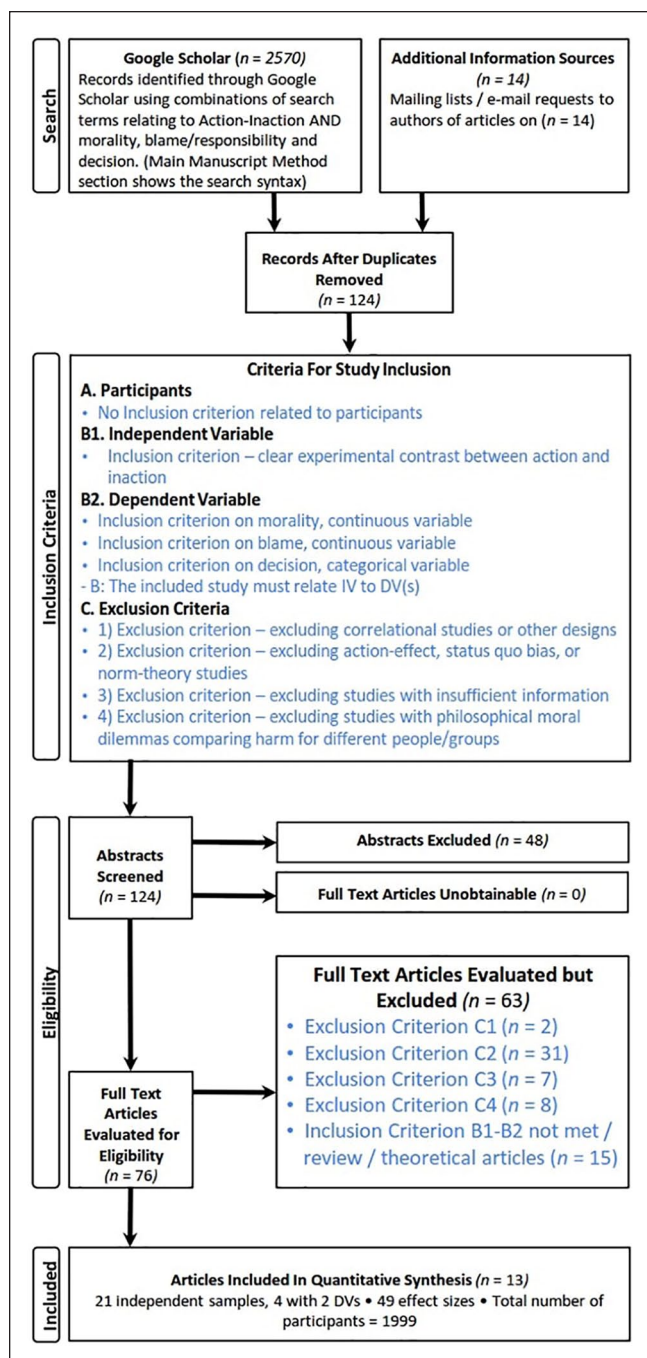


Figure 1. Meta-analysis flow diagram.

Note. We included studies that measured any of the dependent variables. For example, if a study focuses on a decision but does not measure morality/blame, we include the study. We adapted the diagram based on Moreau and Gamble (2020).

(morality OR intent OR harm OR vaccinations OR moral OR blame OR responsibility) – “Trolley.” This process was followed by a scan of the reference sections of found articles. This enabled us to find other articles that were not detected through the search procedure. In addition, we searched for

“related articles” and “cited by” Google Scholar options based on the found articles.

As the last step of this search round, we contacted published authors on the topic and announced our search on ResearchGate, Twitter, and mailing lists asking for published and unpublished to maximize coverage and access to unpublished data and/or manuscripts (Feltz & May, 2017). We conducted this procedure twice, once in 2017 and again in 2020. We received 14 articles previously not found in our search.

As a second step, we scanned all abstracts, tables, and method sections to identify the relevance of the sources. If the articles indicated relevance for our analysis, we read more of the articles to determine whether inclusion criteria were met or whether articles had to be excluded based on our search criteria (see next paragraph). A second scan round enabled us to exclude 48 articles and reduced our sample of studies to 13 articles, including 21 samples. We note that we were surprised by this low number of studies, given how prominent and well-cited the omission bias is in the literature, which is partly the reason for us repeating the search and call procedures. This suggests the need for more studies in this domain. We return to this point in the general discussion.

We listed the included articles in Table 1.

Inclusion and Exclusion Criteria

We established strict inclusion and exclusion criteria to ensure we are capturing omission bias and not any of the other action–inaction effects (see review by Feldman et al., 2020). Excluded articles were noted in the search sheet with exclusion reasons.

We focused on articles referring to omission bias. We excluded articles that were about the action-effect (e.g., Zeelenberg et al., 2000), status quo bias (e.g., Baron & Ritov, 1994), or norm-theory (normality/exceptionality biases, etc.; e.g., Prentice & Koehler, 2002).

Second, we focused on experiments that had the same consequences for both action and inaction, and we only included articles in which the independent variable (IV) had a clear contrast between omission and commission and the dependent variables (DVs) were related to morality (e.g., DeScioli et al., 2011), blame (e.g., Willemsen & Reuter, 2016), or a decision (e.g., Connolly & Reb, 2003) between possible harm through action and possible harm through inaction. This included evaluations of morality wrongness, immorality, moral acceptability, responsibility, blame, and/or moral decisions. We excluded correlational (e.g., Dibonaventura & Chapman, 2008) and one-sample comparison studies (e.g., Zikmund et al., 2006).

Third, the studies had to include adequate statistical information for computing the effect size for a contrast effect between omission and commission. In cases of missing statistical data (e.g., *t* statistics/mean or standard deviation for

Table 1. Studies Included in the Meta-Analysis.

No.	Study	Study/ sample	N	Country	Sample population	Design	Publication status	DV type
1.	Baron and Miller (2000a)	1/1	50	The United States	Students	Within	Published	Decision
2.	Baron and Miller (2000b)	1/2	70	The United States	Students	Within	Published	Decision
3.	Baron and Ritov (2004)	1	112	The United States	General population	Within	Published	Decision, Blame
4.	Bergstrand (2014)	2	115	The United States	Students	Between	Published	Morality, Blame
5.	Blahunka (2014)	1	50	The United States	Students	Within	Not published	Morality
6.	Connolly and Reb (2003)	1	293	The United States	Other	Between	Published	Decision
7.	Cushman and Young (2011)	1	20	The United States	Students	Within	Published	Morality
8.	DeScioli et al. (2011a)	1/1	41	United States	General population	Within	Published	Morality
9.	DeScioli et al. (2011b)	1/2	95	The United States	General population	Within	Published	Morality
10.	DeScioli et al. (2011c)	2	77	The United States	General population	Within	Published	Morality
11.	DeScioli et al. (2012)	1	151	The United States	Students	Within	Published	Morality
12.	Hayashi (2015a)	1/1	80	Japan	Other	Within	Published	Morality
13.	Hayashi (2015b)	1/2	76	Japan	Other	Within	Published	Morality
14.	Hayashi (2015c)	1/3	60	Japan	Other	Within	Published	Morality
15.	Kordes-de Vaal (1996a)	1	74	The Netherlands	Students	Between	Published	Morality, Blame
16.	Kordes-de Vaal (1996b)	3	80	The Netherlands	Students	Between	Published	Blame
17.	Spranca et al. (1991a)	1	38	The United States	Students	Within	Published	Morality
18.	Spranca et al. (1991b)	4	48	The United States	Students	Within	Published	Morality
19.	Willemsen and Reuter (2016a)	2	119	The United States	General population	Between	Published	Blame
20.	Willemsen and Reuter (2016b)	3	116	The United States	General population	Between	Published	Blame
21.	Jamison et al. (2020)	1	313	The United States	General population	Between	Published	Morality, Blame

Note. DV = dependent variable.

calculating Hedge's g , counts, and proportions for omission and commission), we attempted to retrieve the information from authors, and if we failed then we excluded those articles, even if all other search criteria were met. For example, we excluded Haidt and Baron (1996) as they did not report standard deviation, t statistics, and Cohen's d for our effect of interest. Another example is that we excluded Ritov and Baron (1990) because they did not report information relevant for computation of overall omission bias effect size. Their Experiment 1 only reported risk tolerance of different risk levels. Experiment 2 focused on personal decisions versus support for law given different levels of risk. Experiment 4 reported differences between risk cases.

Fourth, we focused on comparisons of morality, blame, and/or decision for a single target and not comparisons of harm inflicted to a different number of people, which confounds moral dilemmas examining utilitarianism. Therefore, trolley dilemmas studies or similar philosophical moral dilemmas were excluded (e.g., Hauser et al., 2007).

Studies collected through the database searches and through contacting authors were assessed for their eligibility based on their titles, abstracts, and contents. One author

determined the adequacy of the study for the meta-analysis and coded the studies, and two other authors verified and adjusted the coding when necessary. All decisions for exclusion were documented and explained.

Coding

We developed a data coding sheet and codebook (see link above). Before we began with the coding process, we pilot-tested six randomly selected studies in two stages and refined it accordingly in every stage. Once the pre-test was completed, we preregistered our plan and proceeded to the full search and coding. One author coded all studies, and two other authors verified and adjusted coding when needed. We reported and justified the decisions clearly.

Analyses

We used RStudio 1.3.1093 Version (RStudio Team, 2020) and metafor package 2.4-0 Version (Viechtbauer, 2010) for the statistical analyses. Given the range of different types of studies and experimental designs, we expected heterogeneity

in the sample to be relatively high. Therefore, we used random effect models.

We converted all effect sizes into Hedge's g to allow for a comparison. We also collapsed split conditions due to moderators in the original studies to allow for a comparison of omission-commission (IVs).

Whenever available, we collected standardized effect sizes directly from authors of original papers. We checked for the accuracy of these analyses based on the provided information and details. If unavailable, we used either descriptive statistics or inferential statistics to re-compute standardized effect sizes. We documented all conversions and coding decisions. We also included the original text in the coding sheet to facilitate reproducibility.

We first conducted a meta-analysis to examine the overall main-effect, for each DV. We conducted two-level models, multivariate three-level models, and MetaForest to examine the impact of the described moderators.

Statistical heterogeneity was determined using the Tau-squared test and quantified using I^2 , which represents the percentage of the total variation in a set of studies that is actually due to heterogeneity (Higgins et al., 2003). This yielded a point estimate, CIs, and p value, along with statistics for heterogeneity, assessed using the Q statistics, and the I^2 statistic. If there was indeed significant heterogeneity, we explored potential moderators.

We conducted several publication bias analyses, examining publication status as a moderator, testing for funnel plot asymmetry (trim and fill, rank test, Egger's unweighted regression symmetry test; Begg & Mazumdar, 1994; Duval & Tweedie, 2000; Sterne & Egger, 2005), and conducted bias assessment and adjustment techniques of P-uniform (van Assen et al., 2015), P-curve (Simonsohn et al., 2014), PET (Precision Effect Test), and PEESE (Precision Effect Estimate with Standard Error (Stanley & Doucouliagos, 2014)).

Data Analysis Strategy

When the project was initiated and preregistered in 2017, we planned to report two-level model results, yet following the completion of the search and coding and recommendations received by open peer review, we decided to adopt multivariate three-level models and moved the two-level model results to the Supplemental Material. We note that the results were very similar.

We used the following strategy for aggregation. If one study with a sample reported two effect sizes of two different DVs, we collapsed those for the main-effect analyses and then analyzed them as separate effect sizes for the separate DV meta-analyses. If one sample reported two effect sizes of the same DV type (e.g., Jamison et al., 2020), we collapsed those into a single effect size. See provided code for more details.

The results for analyses collapsing two effect sizes of two DVs within the sample as one effect size, or analyzing and

treating two or more effect sizes from two or more scenarios within the sample as separate effect sizes, are to a large extent similar. We detail these and discuss the differences in the Supplemental Material (in "additional analyses").

Results

Omission Bias: Overall Main Effect

We conducted the analysis on 21 samples (four with two DVs, the second level in the model), 49 effect sizes, 13 articles (the third level in the model), and found an overall effect size of $g = 0.45$, 95% CI = [0.14, 0.77], with very high heterogeneity. The post hoc statistical power was above 99.99% (Tiebel, 2018). The median power across studies was 66.8% (see Figure 3 metaviz plot for the power distribution of studies; Kossmeier et al., 2020).

We note that we conducted the multivariate three-level model, to account for the dependence (correlations) between effect sizes of the same articles (Cheung, 2019). We did not preregister the use of multivariate three-level analysis, yet adopted the recommendations from an open peer review that the multivariate multilevel model provides a more accurate estimate of the true effect size. The results for the preregistered analysis and the multilevel analysis were very similar, and the analysis for the preregistered analysis was moved to the Supplemental Material.

Omission Bias: Main Effect for Each DV

We proceeded to conduct separate analyses for each of the three DV types. We provide a summary of the results in Table 2. We visualized the results in Figure 2 (forest plots) and Figure 3 (funnel plots).

Studies measuring morality as the DV ($k = 14$) had an overall effect size of $g = 0.71$, 95% CI = [0.47, 0.94]. Studies with blame ($k = 7$) as the DV had an overall effect size of $g = 0.32$, 95% CI = [0.01, 0.64]. Finally, studies examining decision as the DV ($k = 4$) had an effect of $g = 0.30$, 95% CI = [-0.62, 1.21], with CIs overlapping with the null, suggesting weak to no support for omission bias.

Following up with an unregistered exploratory analysis, we excluded Connolly and Reb (2003), which was flagged as an outlier, as an article showing a general tendency toward commission. When excluding this article, we found a slightly larger overall effect size of $g = 0.56$, 95% CI = [0.32, 0.80], and a more consistent and larger effect size for decision as a DV $g = 0.71$, 95% CI = [0.16, 1.25].

Heterogeneity

Using a multivariate three-level model, we conducted a Cochran's Q test to check whether variations in the observed effect size were likely to be attributable solely to sampling error. The variation in effect size was greater

Table 2. Meta-Analytic Results for Dependent Variables of the Omission Effect.

Measure	<i>k</i>	<i>Q</i>	<i>df</i>	I^2 [%]
Morality	14	56.56	13	79.72
Decision	4	66.39	3	94.69
Blame	7	26.51	6	82.31

Note. *k* = number of samples; *Q* = Cochran's *Q* test for heterogeneity; I^2 = I square.

than what would be expected from sampling error alone $Q(48) = 271.22, p < .001$, indicating a variation of true effect across studies.

Moreover, we computed I^2 and found high heterogeneity between the studies ($I^2 = 93.68\%$, 75% is regarded as high variability; Huedo-Medina et al., 2006). We reported the heterogeneity statistics of all three DVs in Table 2.

Excluding Connolly and Reb (2003), the heterogeneity was lower compared with Connolly and Reb (2003) but still substantial, $Q(47) = 180.04, p < .001, I^2 = 84.24\%$.

Publication Bias

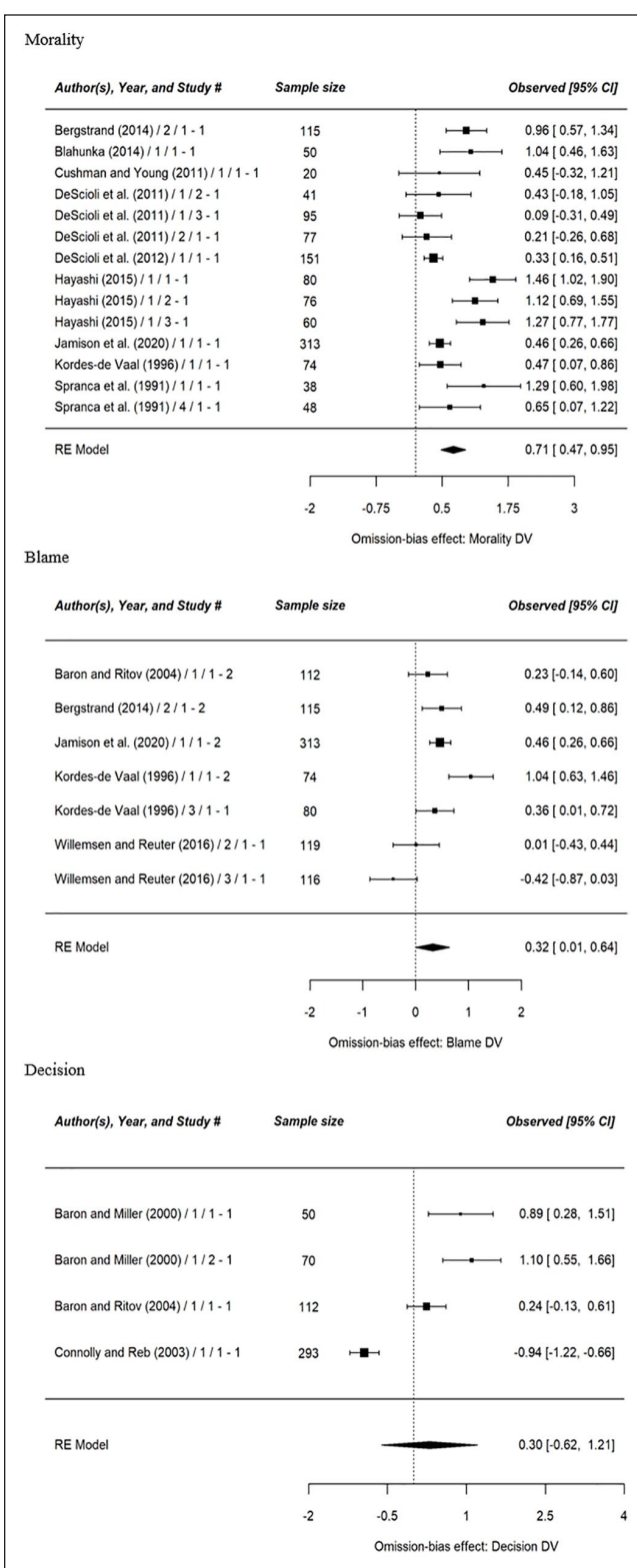
We used several statistical approaches to examine publication bias. The omission bias using adjustment methods of P-uniform, P-curve, and Henmi and Copas (2010) showed effects meaningfully different from the null. However, the adjusted effect using the three-parameter selection model (Iyengar & Greenhouse, 1988) was $g = 0.34$, 95% CI = $[-0.03, 0.72]$, with the effect overlapping with the null, with weak to no omission bias effects using PET and PEESE techniques.

We provide funnel plot and asymmetry tests, including Egger's regression test and Rank correlation test, in Tables 3 and 4, and Figure 3, showing several *p* values close to .05, slightly below or above, which with the publication bias assessment are indicative of possible publication bias. We discuss publication bias results in more detail in the section "Discussion."

We also preregistered and conducted a moderator analysis comparing the effects for published versus unpublished studies. However, we were only able to identify one unpublished study, and so the analysis is reported in the Supplemental Material (section "published vs. unpublished moderator analysis").

Moderators Analysis

We initially adopted fixed-effects two-level models to contrast effects of moderators' categories and conducted multivariate three-level models to account for confounding relationships and dependence of effects within the same article. However, due to the small number of studies included we realized we have insufficient power to draw conclusions from these moderator analyses. Therefore, following peer

**Figure 2.** Meta-analysis forest plots—Morality, blame, and decision.

review advice, we supplemented our analyses with MetaForest moderator analyses (Curry et al., 2018; van Lissa, 2017), which applies bootstrapping, to address the

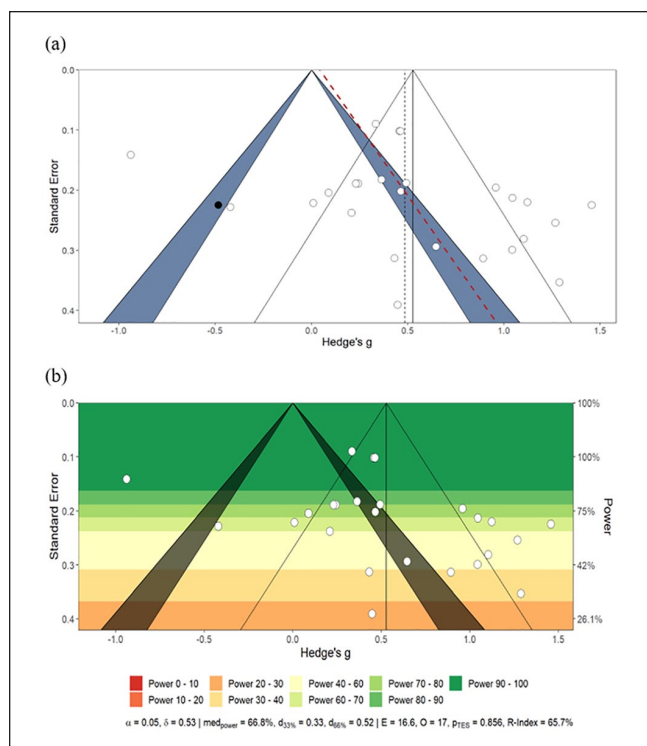


Figure 3. Meta-analysis funnel and sunset plots.

Note. Created using metafor (Viechtbauer, 2010) and metaviz (Kossmeier et al., 2020) R packages. The blank dots of the trim-and-fill funnel plot refer to the actual observed effects of different studies whereas the black dot refers to the effect of the missing imputed additional study to ensure the plot is symmetric. The different colors in the sunset plot refer to different levels of statistical power.

issue of limited statistical power as each moderator category has limited studies. We report the underpowered two- and three-level moderator analyses in the Supplemental Material (section “two-level and multivariate three-level moderators analyses”).

First, the key indicator of the MetaForest model— R^2 (R-OOB) was .29, meaning that the moderators predicted variance in the effect ($I^2 = 84.56\%$, $Q = 267.61$, $p < .001$). Furthermore, we found that all five moderators had positive variable importance, with responsibility for target being the most important variable (stronger effects for no clear responsibility for target studies compared with clear responsibility for target studies), followed by presence/absence of negative outcome information, self versus other, familiarity with target, and design type. See Figure 4.

Familiarity with the target. Twelve studies for familiarity with the target had an effect size of $g = 0.58$, 95% CI = [0.24, 0.91]. Six studies in which the decision maker was not familiar with the target showed an effect of $g = 0.28$, 95% CI = [0.14, 0.41]. We failed to find support for the hypothesized difference between more familiar with target and less familiar with target studies with fixed-effects model and multivariate three-level model. Using MetaForest, familiarity with

target was the second least important variable, with positive variable importance. We conclude weak support for this moderator, though not in the expected direction.

Role responsibilities for targets. Seven studies that employed scenarios with clear role responsibilities for targets had an effect size of $g = -0.05$, 95% CI = [-0.42, 0.32]. Seventeen studies with studies that had no clear responsibility over the target showed an effect size of $g = 0.60$, 95% CI = [0.44, 0.76]. Using MetaForest, responsibility over target was the most important variable. The results indicate support for the hypothesis that there is stronger omission bias when people have no clear social role responsibility, compared with when they are in a social role with a clear responsibility.

Self versus others. Four studies with self as the target had a mean effect of $g = 1.42$, 95% CI = [0.60, 2.24], whereas the 25 studies with others as targets had a much weaker effect of $g = 0.46$, 95% CI = [0.26, 0.65]. With MetaForest, self versus others was the third most important variable in predicting the effect.

Outcome. Thirteen studies with negative outcomes had a mean effect of $g = 0.71$, 95% CI = [0.42, 1.01]. Thirteen studies with no clear negative outcome or neutral outcome also had a weaker effect size of $g = 0.30$, 95% CI = [0.02, 0.58]. With MetaForest, the presence/absence of clear negative outcomes was the second most important variable. This indicates that it is likely there would be a stronger effect for studies with negative outcomes compared with studies with no clear negative outcome.

Study design. We analyzed nine between-subject studies and found an effect size of $g = 0.43$, 95% CI = [0.17, 0.70]. Sixteen within-subject studies had a mean effect size of $g = 0.59$, 95% CI = [0.28, 0.91]. Using MetaForest, even though study design was the least important variable, it has positive variable importance. We conclude weak support for this moderator, though in the expected direction.

Discussion

Omission Bias: Summary of Findings

We provided a summary of our meta-analysis findings in Table 5. The meta-analysis findings were in support of an omission bias with a mean effect of $g = 0.45$, 95% CI = [0.14, 0.77], though we note there were indications of a publication bias and the number of studies we were able to identify was surprisingly low.

This suggests one of several insights. First, despite three decades of research on omission bias and being a well-cited and reputable phenomenon, we required more research to allow for stronger conclusive evidence for omission bias, and to address the possible issues of publication bias, using solutions such as registered reports, crowdsourced replications, and/or adversarial collaborations.

Table 3. Publication Bias Adjustment.

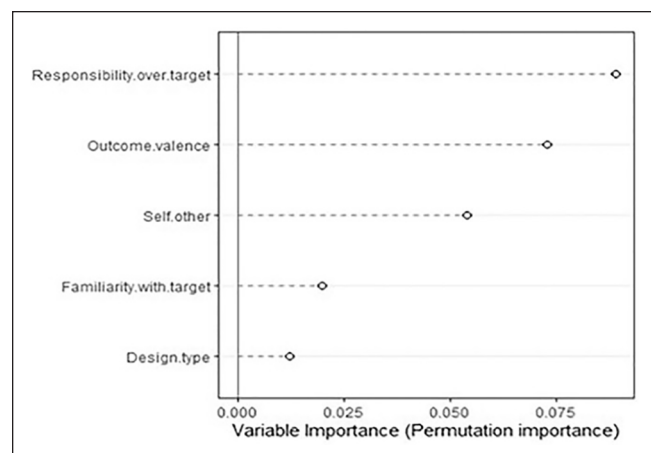
Adjustment Method	Hedge's <i>g</i>	CI
Trim and Fill (Duval & Tweedie, 2000)	0.49	[0.26, 0.72]
Three-parameter selection (Iyengar & Greenhouse, 1988)	0.34	[-0.03, 0.72]
Henmi and Copas (2010)	0.41	[0.12, 0.70]
P-uniform (van Assen et al., 2015)	0.74	[0.48, 1.00]

Note. Analyses conducted on multivariate three-level model: $k = 49$, $g = 0.45$, 95% CI = [0.14, 0.77], before correction. CI = confidence interval.

Table 4. Other Publication Bias Analyses.

Publication bias analysis method	Results
Precision Effect Test (Stanley & Doucouliagos, 2014)	$b = 0.04$, 95% CI = [-0.49, 0.57]
Precision Effect Estimate with Standard Error (Stanley & Doucouliagos, 2014)	$b = 0.21$, 95% CI = [-0.11, 0.53]
Rank correlation test (Begg & Mazumdar, 1994)	Kendall's tau = .20, $p = .170$
Sterne and Egger Regression test (2005)	$z = 1.97$, $p = .048$
P-curve (Simonsohn et al., 2014)	Evidential value is present, adjusted effect: $d = 0.70$

Note. Values in parentheses indicate 95% confidence intervals [lower bound, upper bound].

**Figure 4.** MetaForest variable importance plot.

Note. See the rest of MetaForest analyses in the Supplemental Material. The higher the variable importance, the stronger the moderating effects.

Second, our moderator analyses suggest that omission bias is affected by several moderators, raising the need for a follow-up systematic investigation of the promising moderators we identified and tested in this meta-analysis. We discuss those in more detail below.

Furthermore, we found that the omission effect was meaningfully different from null for morality and blame as DVs. The effect CIs for decision overlap with null.

Omission Bias: Effects of Different DVs

We found support for omission effect in third-person morality ratings and blame of others' behavior yet failed to find support for a reliable effect for decisions (CIs overlapped with the null). The lacking support for decisions may be

attributed to the following: (a) low power with only four samples, (b) findings in the opposite direction in the context of vaccination (Connolly & Reb, 2003).

We therefore note caution regarding drawing any strong conclusions regarding the effect of decision based on the limited set of studies included. It appears that the effect of omission bias in first-person decision-making is less reliable and less consistent, as our decisions are related but not simply influenced by moral judgments. We are unable to draw strong inferences regarding the causes or mechanisms regarding decisions based on our small sample of included studies (causal chain being explicit, Baron & Ritov, 2004; vaccination context, Connolly & Reb, 2003; differences between decisions and judgments). More research is needed to address this debate, and we see special value in conducting a large-scale investigation of this research question, preferably involving an adversarial collaboration in the form of a registered report.

Moderators

Familiarity with target. Previous studies (e.g., Baron & Ritov, 2004; Haidt & Baron, 1996) demonstrated that familiarity with the target affects omission bias. Specifically, omission effect was weakened when actors had high solidarity with the target (Haidt & Baron, 1996). We refrain from drawing strong conclusions based on such uncertain null findings, yet identified differences in the opposite direction may be due to some unknown confounding variables or artifacts of some experiments. Another possible explanation for the confusing findings is that in Haidt and Baron (1996) study Experiment 1 and Experiment 2, the affected targets are friends but not family members. We included studies in which affected targets are friends but also studies in which affected targets are

Table 5. Summarized Results of the Meta-Analysis.

Measure/ Moderator	Hypothesis	Result
Morality, blame, and decision as DVs	H1a for morality: People judge harm through action as less moral, compared with harm through inaction.	H1a for morality: Supported. $g = 0.71$, 95% CI = [0.47, 0.94]
	H1a for blame: People judge harm through action as more morally responsible/blameworthy, compared with harm through inaction.	H1a for blame: Supported. $g = 0.32$, 95% CI = [0.01, 0.64]
	H1b: People prefer inactive decision over active decision given a possibility of harm.	H1b: Not supported. $g = 0.30$, 95% CI = [-0.62, 1.21]
Familiarity with target	H2a: Omission bias is stronger for more familiar targets H2b: The omission bias effect is stronger for less familiar targets (based on Haidt & Baron, 1996; not preregistered)	H2: No evidence for H2b, weak evidence for H2a.
Responsibility for target	H3a: The omission bias effect is stronger the higher the perceived responsibility for the target. H3b: The omission bias effect is stronger given unclear or lower social role responsibility for the target.	H3b: Supported.
Self–other	Exploratory not preregistered: The omission bias effect is stronger when the target is self, compared with when the target is others.	Support for moderation.
Negative outcome information	H4: There is an omission bias effect even when outcomes are not negative (null not included in confidence interval). H5: The omission bias effect is stronger when outcome is negative compared with when it is not negative (neutral or no information).	H4: Supported. H5: Mixed support, likely moderation. Supported in MetaForest and the two-level model, not supported in three-level.
Study design	Competing H6a: The omission bias will be stronger for studies using a within-subject design. Competing H6b: Study design will have no impact on the omission bias.	H6a: Mixed support, likely moderation. Supported with MetaForest, not supported in the two-level or the three-level models.

Note. “likely moderation” concluded based on MetaForest, which uses bootstrapping to compensate for low power. DV = dependent variable.

family members (e.g., children or spouse). It is possible that the weakening of omission bias is only constrained to situations in which the affected targets are friends, but not family members. It is also possible that the role of familiarity in omission bias is minimal, or in the opposite direction to Haidt and Baron’s (1996) findings. We believe there is value in conducting more well-powered replications of the classic findings in omission bias (notably Haidt and Baron, 1996) with extensions testing different levels of familiarity.

Responsibility over target. The findings support the hypothesis, based on Haidt and Baron (1996), that omission bias is weakened given clear social role responsibility, in contrast to situations in which the decision maker has no clear social role responsibility. This may be because when people have no clear responsibility for the target, they are not expected or required to act, thereby exhibiting omission bias, which means they are more likely to choose inaction or judge inaction less harshly than action.

Target: self versus others. We found support for omission bias regardless of whether the affected target was not the self, different from the findings of Zikmund-Fisher et al. (2006). They found that most participants prefer active treatments when the affected target is others. Moreover, our findings

that omission bias is stronger when affected target is self is consistent with studies in medical decision-making (Janssen et al., 2015; Mendel et al., 2010; Ubel et al., 2011), which do not specifically contrast action versus inaction. However, we note that the number of studies where self is the target was very limited, with only four samples from two articles in our meta-analysis. We call for more studies, perhaps through direct and conceptual replications and more follow-up studies with self as target and others as target to compare. For example, in the context of a pandemic, parents need to decide whether they would vaccinate themselves, and decide for their children as well. Furthermore, more work is needed to investigate the purposed mechanisms and factors (anticipated emotions, social role responsibility, cost-benefit analysis) if such effect is meaningful.

Presence/absence of negative outcome. Our findings indicated support for omission bias regardless of outcome valence, which is consistent with a recent replication (Jamison et al., 2020) of Spranca et al. (1991), and supported the notion that omission bias is stronger when the outcome is negative. In real-life, as outsider observers, we are more likely to be informed about the decision if the outcome is negative compared with when the outcome is neutral. This may imply outcome bias in omission bias occur frequently in real-life

settings, from daily life judgments to judgments in legal settings. Our findings are in line with findings of outcome bias in ethical judgments (Gino et al., 2009).

Before making a decision, there is no way of knowing for sure if the outcome will be negative, neutral, or positive. This may partly explain why omission bias findings are inconsistent and may be weakened or even reversed into general tendencies toward action, compared with outsider judgments of omission versus commission.

More studies using different moral scenarios with different measures, including binary decisions and third-person judgments, under different kinds of outcome information (positive outcome, no outcome information, neutral outcome, negative outcome) are needed to verify such notions and compare the effects more systematically in a well-controlled manner.

Study design. In the domain of action-effect research, some (e.g., Connolly & Reb, 2001; N'gbala & Branscombe, 1997) have found that between-subject studies did not replicate the within-subject findings in Kahneman and Tversky's (1982) investor scenario successfully. We found meaningful effects even using between-subject design, though we note mixed support for differences in effect size between designs. We call for more research to adopt both within design and between design (e.g., Anvari et al., 2021; Ziano, Kong, et al., 2021; Ziano, Xiao, et al., 2021) compare effects in different settings, and comparing effects of judgment, emotions, and decisions of action–inaction-related effects using joint evaluation versus separate evaluation.

Both within and between designs have their merits. Jamison et al. (2020) adjusted to between-subject design as it can test the generalizability of omission bias to everyday life situations, with incomplete information regarding behavior and outcomes, as most times people do not witness all possibilities of a situation. Spranca et al. (1991) major rationale behind using a within-subject design was to hold intentionality constant, as in a between-subject design, people may infer differently about intentionality of commission and omission, as pointed out by Jonathan Baron in his peer review of Jamison et al. (2020). See Table 6 for a comparison of within-subject studies versus between-subject studies.

Reversal to Commission Effect⁴ in Some Studies? Context, Expectations, Justifiability, Normality, and Individual Differences

Even though our study found an overall effect of omission bias, omission bias is not universally applicable to all contexts and all individuals. As mentioned above, omission bias was weakened by several moderators.

Furthermore, preferences may shift toward commission under some circumstances, a point that both Baron and Ritov (2004) and Connolly and Reb (2003) seemed to agree on. Connolly and Reb (2003) found that the majority of participants preferred vaccination versus non-vaccination, using

choice measures. This is consistent with Zikmund-Fisher et al. (2006), which also used choice measures, yet contradicts most omission bias findings in vaccination that used risk-balancing procedures (e.g., Asch et al., 1994; Ritov & Baron, 1990), and also choice measures (e.g., Baron & Ritov, 2004). Moreover, Connolly and Reb (2003) argued that general tendencies for action versus inaction have limited generalizability to decisions regarding vaccinations. The method (binary choice vs. risk balancing procedures) and context (vaccination) may influence whether omission bias or general tendencies toward commission is found. There may also be temporal and cross-cultural constraints of generality (Simons et al., 2017) regarding vaccination (e.g., Seo & Lim, 2018; Walsh et al., 2016). Unfortunately, in the recent decade, there are very limited studies tying vaccination decisions with omission bias, which is a great loss given how important this phenomenon is in the ongoing pandemic.

The context may affect whether it is actions or inactions that are more justified and socially expected. If an action is justified (Connolly & Reb, 2003) and expected (Bar-Eli et al., 2007; Baron & Ritov, 2004; Ritov & Baron, 1994), then there may be stronger preferences toward taking action than inaction. There is much promise in further studies on action versus inaction decision in moral scenarios looking at morality measures, and examining moderators regarding the role of justifiability, expectations, and normality.

Whether an action or an inaction is justified is associated with whether the decision maker expects to feel regret for an action or an inaction (Connolly & Zeelenberg, 2002). Individual differences studies in omission bias/action bias sometimes have so far yielded inconsistent results (e.g., Chung et al., 2014), and sometimes even contradictory findings (Ritov & Baron, 1999, vs. Tanner & Medin, 2004). We believe that there is much need of revisiting these findings, attempting to resolve such discrepancies, and investigating the role of individual differences in the omission bias.

Implications, Limitations, and Directions for Future Research

Our meta-analysis is not without limitations. Below, we discuss some of the issues encountered while conducting the meta-analysis.

First, during the meta-analysis coding process, we experienced difficulties in analyzing statistical information to calculate effect sizes. We had to exclude several studies due to insufficient statistical information. In addition, we excluded studies that had no clear contrast between action and inaction. As a result, the number of included studies in the meta-analysis is relatively low. Despite the tremendous implications of omission bias in real-life decision-making, especially in health-medical settings, there are unfortunately very limited studies testing binary decisions with clear action–inaction contrasts. We call for more studies testing binary decisions with the action–inaction framework.

Table 6. Comparison Between Within-Subject Studies and Between-Subject Studies.

Issue	Within-subject design	Between-subject design
Usage	All participants compare or evaluate two options or decisions displayed together.	Participants are randomized into two or more conditions. One group evaluates one option or one decision whereas another group evaluates a different option or decision.
Example in omission bias	Evaluating morality ratings of both omission and commission (Spranca et al., 1991; Study 1). Omission refers to not saying anything to prevent despite knowing the tennis match opponent is allergic to the salad. Commission refers to recommending the tennis match opponent to eat the salad which he would be allergic to. $g = 1.29$, 95% CI = [0.60, 1.98], larger effect.	Evaluating morality ratings of either omission or commission (Jamison et al., 2020; Scenario 1; replication of Spranca et al., 1991; Study 1). $g = 0.46$, 95% CI = [0.26, 0.66], smaller effect.
Merits	1. Control for contextual factors or other confounding variables, such as intention in omission bias 2. Higher statistical power	1. More generalizable to real-life situations where people only evaluate one of the options or one of the decisions. We may not be exposed to all possibilities. 2. Reduces risks of demand effects
Key issues or debates	1. What effects demonstrated in within-subject design are replicable in a between-subject design? More replications with methodological adjustments of action–inaction literature are needed. 2. What phenomena show weaker effects with between-subject design compared with within-subject design? Future omission bias articles, and other judgment and decision-making research, can include both within-subject design and between-subject design. 3. Are there/how large are demand effects using within-subject designs? Charness et al. (2012) argued that demand effects are more likely using within-subject designs. However, some studies (e.g., Lambdin & Shaffer, 2009; Mummolo & Peterson, 2019) found that participants are very unlikely to guess the hypotheses of within-subject studies correctly or very unlikely to adjust their behavior to fit the researchers' expectations. Future omission bias studies should ask participants if they can guess the purposes or hypotheses and exclude participants who guess correctly or measure demand effects (De Quidt et al., 2018).	

Second, the number of studies included in the meta-analysis was too small to reliably test moderators using traditional moderator analyses, likely leading to less accurate effect size estimates.

Our findings using the different moderator methods were mixed for some of the moderators. Given the low power, we consider MetaForest as the most appropriate. We refrain from drawing strong conclusions regarding some analyses (especially regarding decision as DV and study design moderating effect). We hope to see more work done in this domain, hopefully employing large samples and data sharing, conducting replications to revisit classic omission bias findings, and using preregistrations and registered reports format. Furthermore, given the debates over this effect, we see great value in adversarial collaborations (see, for example, Bateman et al., 2005; Mellers et al., 2001). These may help address some of the inconsistencies and contradicting findings in the omission bias literature, such as the vaccination decision debates between Connolly and Reb and Baron and Ritov. Such inconsistencies in findings, sometimes with findings in opposite direction, may be due to differences in methods (binary choice vs. risk balancing methods). Furthermore, regarding the debates about study design (within-subject design vs. between-subject design), adversarial collaborations may be helpful in resolving disagreements and gaining a better understanding of the differing perspectives.

Finally, the vast majority of included studies were from Western, Educated, Industrialized, Rich and Democratic (WEIRD) countries (Henrich et al., 2010), especially the United States (Cheon et al., 2020), except Hayashi (2015) in Japan as well as Baron and Miller (2000) in India. There have been some cross-cultural studies in action–inaction attitudes and goals (Ireland et al., 2015; Zell et al., 2013) and action-effect (Chen et al., 2006; Gilovich et al., 2003), but less have been done cross-culturally on the related omission bias focusing on action and inaction in domains of morality and harm. We call for more cross-cultural multi-lab collaborations to test the generalizability of omission bias (such as psychological science accelerator, see Moshontz et al., 2018).

In summary, we call for conducting more studies on omission bias. Further studies are needed to better understand action–inaction decisions in situation involving possible harm, especially examining important real-life implications such as vaccination decisions. Future research may follow with a systematic investigation of moderators such as expectations, justifiability, normality, length of casual chain, individual and cultural differences, and examining robustness and generalizability using different methods, measures, and contexts.

Conclusion

We conducted a meta-analysis and found support for an omission bias yet note possible publication bias. Our

moderator analyses showed strong support for responsibility over target, and mixed/some support for outcome, self versus other, and design type as moderating variables of the omission bias. The number of studies we found was small, and so we hope that this meta would drive further research in this domain, and call for more preregistered replications and extensions and follow-up work. We also call for adversarial collaborations using registered reports to help resolve mixed findings in the literature.

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Author Contributions

Tijen Yay worked under the supervision of Gilad at Maastricht University for conducting the preregistered meta-analysis as part of her masters' thesis. Tijen wrote the preregistration, with verification and registration by Gilad. Tijen conducted the search of the literature. Gilad developed the coding scheme. Tijen and Gilad pre-tested the coding sheet. Tijen and Gilad coded the articles. Gilad wrote the RMarkdown code and analyses. Tijen summarized the methods and results and wrote an initial draft. Siu Kit verified coding, analyses, and results, made modifications to the code, and rewrote the manuscript to prepare for journal submission. Siu Kit wrote the final journal submission draft, with Gilad guiding and editing.

Contributor Roles Taxonomy (Allen & O'Connell, 2014)

Role	Siu Kit Yeung	Gilad Feldman	Tijen Yay
Conceptualization		✓	✓
Pre-testing		✓	✓
Preregistration		✓	✓
Data curation	✓	✓	✓
Formal analysis	✓	✓	✓
Funding acquisition			
Investigation	✓	✓	✓
Preregistration peer review/verification		✓	
Literature search	✓	✓	✓
Datafile study/effect coding	✓	✓	✓
Reproducible code (e.g., RMarkdown)	✓	✓	✓
Contacting authors		✓	✓
Data analysis peer review/verification	✓		

(continued)

Role	Siu Kit Yeung	Gilad Feldman	Tijen Yay
Methodology	✓	✓	✓
Project administration		✓	
Resources			
Software	✓		✓
Supervision		✓	
Validation	✓		
Visualization	✓		✓
Writing-original draft		✓	✓
Writing-review and editing	✓	✓	

Declaration of Conflicting Interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: The authors note that Jamison et al. (2020), one of the included articles, includes two authors of this meta-analysis—Tijen Yay and Gilad Feldman.

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ORCID iD

Gilad Feldman  <https://orcid.org/0000-0003-2812-6599>

Supplemental Material

Supplemental material is available online with this article.

Notes

1. Risk balancing procedure refers to the procedure in which the experimenter only provides the probability of omission leading to harm and asks the participants the probability of commission (e.g., vaccination) causing harm, which would make both options indifferent. Check Connolly and Reb's (2003) appendix for examples. Our meta-analysis focuses on morality, judgments, and decision as dependent variables (DVs) but not risk-premium.
2. H2a and H3a: We preregistered this hypothesis, as a result of an initial misunderstanding regarding Haidt and Baron (1996), check Supplementary "Preregistration Versus Final Report Deviations" (pp. 20–21) for details.
3. H2b and H3b: Based on a revised understanding of Haidt and Baron (1996), they were added as counter to H2a and H3a after preregistration. H2a/H2b and H3a/H3b test the same moderation, though framed differently. Both H2a/H3a and H2b/H3b are included to reflect the process that we went through between preregistration and the final write-up.
4. We note that the preference for commission instead of omission may be beneficial in some contexts (in those contexts, the term "effect" may be more appropriate), but may have disadvantages in some contexts (in those contexts, that may be a "bias"). However, this is not a major objective of our meta-analysis.

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